

David Johnston

Career Objective

Senior Data Engineer with 20 years in Data Analysis, currently at Nova Tech Group. I work at the point where architecture meets delivery: setting technical direction, then staying close enough to the code to know whether it held. Strongest in NumPy, Python and PostgreSQL.

Experience

Senior Data Engineer

Nova Tech Group | Sep 2013 – Now

- » Designed the experiment readouts template now used company-wide.
- » Rebuilt the finance forecast with the FP&A; team, improving accuracy by 15%.
- » Analysed churn drivers and informed retention work that reduced churn by 40%.

Data Analyst

Silverline Data | Sep 2007 – Aug 2013

- » Built Power BI dashboards used daily by 10 stakeholders across three departments.
- » Built data quality tests in Spark that catch upstream breakages before stakeholders see them.

Northbridge IT — Business Intelligence Analyst

Sep 2006 – May 2007

- » Modelled pricing scenarios in Airflow, supporting a 12% margin improvement.

Junior Data Analyst

Summit Digital | Sep 2005 – Sep 2006

- » Trained 15 non-technical colleagues to self-serve, cutting ad-hoc request volume by 12%.

Education & Qualifications

PhD, Mathematics

Michaelfurt State University | 2000 – 2005

Thesis: "Convergence bounds for stochastic optimisation methods"

Bachelor, Mathematics

Shannonshire State University | 1997 – 2000

Core Competencies

Languages: Python, Java

Frameworks: NumPy, PyTorch, Pandas

Cloud & DevOps: Kubernetes

Data: PostgreSQL, Airflow, Snowflake, BigQuery, dbt, Spark, Redshift

Analytics & Design: Excel, Power BI

Testing & Tooling: Playwright

Personal Projects

Home energy dashboard

Collects meter readings and renders them with NumPy. Mostly an excuse to learn time-series storage properly.

Personal finance tracker

Self-hosted spend tracker built with Redshift. Deliberately kept the data local rather than using a hosted service.

Training & Certifications

Databricks Certified Data Engineer Associate — Databricks, 2023

Azure Solutions Architect Expert (AZ-305) — Microsoft, 2019

ISTQB Certified Tester - Foundation Level — ISTQB, 2022

Publications

"Graph partitioning under capacity constraints" - technical report, 2006. Cited internally as the basis for the current design.

Hobbies & Interests

Cycling, Five-a-side football, Open-source contribution